Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 01** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Case Study based question of 4 marks with internal subparts.
6. Use of calculators is not permitted. All rough work and calculation steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. For a given frequency distribution, the class mark of the class interval $15 - 35$ is: [1]
(A) 20 (B) 25 (C) 30 (D) 15

Q2. While computing the mean of grouped data, it is assumed that the frequencies are: [1]
(A) centred at the upper limits of the classes
(B) centred at the lower limits of the classes
(C) centred at the class marks of the classes
(D) evenly distributed across all classes

Q3. If $\sum f_i = 20$ and $\sum f_i x_i = 200 + 5k$, and the mean of the distribution is 15, then the value of k is: [1]
(A) 15 (B) 20 (C) 25 (D) 30

Q4. The modal class for the following distribution is: [1]

Class	0–10	10–20	20–30	30–40	40–50
Frequency	7	14	13	12	20

(A) $10 - 20$ (B) $20 - 30$ (C) $30 - 40$ (D) $40 - 50$

Q5. **Assertion (A):** The mean of the first 10 prime numbers is 12.9.
Reason (R): The first 10 prime numbers are 2, 3, 5, 7, 11, 13, 17, 19, 23, and 29. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Find the mean of the first 6 multiples of 3. [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. Find the arithmetic mean of the following distribution using the Direct Method: [3]

Class Interval	0–10	10–20	20–30	30–40	40–50
Frequency (f_i)	7	10	15	8	10

Q8. Find the mode of the following grouped frequency distribution: [3]

Class	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	4	8	10	12	6	4

Q9. Compute the median of the following frequency distribution: [3]

Class Interval	0–20	20–40	40–60	60–80	80–100	100–120
Frequency	6	8	10	12	6	5

SECTION D: Case Study Based Question (1 × 4 = 4 Marks)

Q10. CASE STUDY: HOSPITAL INPATIENT COVID-19 CLINICAL ADMISSIONS PROFILE [4 Marks]

A municipal medical hospital recorded the age distribution of patients admitted during a seasonal influenza and respiratory illness surge to allocate medical oxygen, intensive care beds, and nursing personnel efficiently.

Age Interval (years)	5–15	15–25	25–35	35–45	45–55	55–65
Number of Patients	6	11	21	23	14	5

(i) Identify the modal class of the admitted patients and state its lower limit. [1]

(ii) Calculate the mode of the age distribution of the patients. [2]

(iii) Find the class mark and cumulative frequency of the modal class. [1]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★